

HACKATHON PROJECT GUIDE

GEOSPATIAL AI — SATELLITE-DRIVEN PRECISION AGRICULTURE

AI-DRIVEN AUTOMATED CROP TYPE, MOISTURE
STRESS
DETECTION & IRRIGATION ADVISORY SYSTEM

Across Growth Stages Using Moderate-Resolution Optical & Microwave Satellite
Data

TEAM NAME

MeraCodeLikhDo

Team Member	Role
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1. Introduction & Background

Indian agriculture depends heavily on the timely availability of water, and the gap between crop water demand and actual water supply is one of the biggest drivers of yield loss, farmer distress and inefficient canal-water usage. At the same time, national missions such as PMKSY, the Digital Agriculture Mission, NMSA and PMFBY are increasingly looking for scalable, satellite-driven intelligence rather than slow, manual field surveys.

Earth observation has matured to a point where optical sensors (LISS-III, AWiFS, Sentinel-2, Landsat, MODIS) and microwave SAR sensors (EOS-04, Sentinel-1, and the upcoming NISAR mission) can be fused to deliver continuous, all-weather monitoring — something that is critical during the cloud-heavy kharif/monsoon season when optical data alone becomes unusable for weeks at a stretch.

This project proposes an automated, AI-driven pipeline that goes beyond simple vegetation-index monitoring. It identifies crop type, tracks moisture stress in a phenology-aware (growth-stage-specific) manner, and converts that stress signal into a practical, field-level irrigation advisory — closing the loop from raw satellite pixels to an action a farmer or canal-command operator can actually use.

2. Problem Statement

Despite the availability of rich satellite archives, several practical gaps remain in how moisture stress and irrigation needs are currently assessed:

- Cloud cover during the kharif/monsoon season frequently blocks optical satellites for weeks, leaving large monitoring gaps when crops are most vulnerable to water stress.
- Most existing stress-detection approaches rely on simple NDVI thresholds and ignore the fact that the same index value means something different at sowing, tillering, flowering or maturity — i.e. they are not phenology-aware.
- Crop-type maps are often produced late in the season or require extensive, costly ground surveys, limiting their use for early-season planning.
- Irrigation scheduling on the ground is frequently based on convention or guesswork rather than a quantified 8-day water deficit, leading to either water wastage or avoidable crop stress.
- Crop insurance verification under PMFBY lacks a continuous, auditable, satellite-backed evidence trail of stress and damage progression.

This project addresses each of these gaps within a single, fusion-based, end-to-end pipeline.

3. Project Objectives

1. Develop an automated AI/ML methodology for crop type identification for the current cropped season, using multi-temporal spectral signatures learned from the previous cropped season.

2. Detect moisture stress dynamics across crop growth stages by integrating phenology-aware modelling with optical and microwave indicators, enabling stage-wise stress interpretation.
3. Build a satellite-based system that estimates 8-day crop water deficit and generates field- and command-area-level irrigation advisory maps.

4. Unique Selling Points (USP)

Several teams at any geospatial hackathon will build an NDVI-threshold stress map. The points below are what are intended to differentiate this submission — each one targets a real adoption barrier in operational agriculture monitoring, not just a modelling trick.

4.1 Foundation-Model-Powered Few-Shot Crop Classification

Instead of relying purely on hand-engineered spectral indices and large labelled datasets, the pipeline plugs in pretrained Earth-observation foundation model embeddings (e.g. Google DeepMind AlphaEarth Foundations, NASA-IBM Prithvi-EO-2.0, or the Clay Foundation Model). These embeddings already encode global land-cover and crop-phenology structure, so a lightweight classifier on top can reach strong accuracy with far fewer ground-truth points — directly solving the “ground truth is scarce” problem that limits most operational crop maps.

4.2 True Stage-Level Optical–SAR Fusion, Not Just Gap-Filling

Most fusion pipelines use SAR only to patch holes left by clouds. This design fuses VV/VH backscatter and texture with optical indices at the level of each phenological stage (sowing, vegetative, flowering, maturity), so the stress index itself is stage-aware rather than a single blended number — giving agronomically meaningful, not just statistically smoothed, output.

4.3 Confidence-Aware Advisory Engine

Every irrigation advisory ships with an uncertainty flag (via Monte Carlo dropout / quantile estimation), so a canal operator or farmer can see when the model is confident versus when ground verification is recommended. This single feature is what separates a research demo from something an operator would actually trust enough to act on.

4.4 Farmer-First Multilingual Delivery Layer

A dashboard alone rarely reaches the field. An LLM-based natural-language layer converts the numeric deficit/advisory output into short, plain-language SMS, WhatsApp or IVR voice messages in regional languages (Hindi and others), closing the last-mile gap between a geospatial model and an actual farmer decision.

4.5 Canal Command Water-Budget Optimizer

Beyond visualizing deficit, the system aggregates pixel-level demand up to the canal outlet / command-area level and suggests a relative water-release prioritization across outlets — turning a monitoring map into an operational scheduling input for irrigation departments.

4.6 NISAR-Ready, Future-Proof SAR Ingestion

The SAR ingestion module is built as a swappable adapter so that once NISAR data streams become available, the same pipeline can ingest higher-resolution L-band/S-band backscatter

with minimal rework — a deliberate first-mover design choice rather than a one-off Sentinel-1 integration.

4.7 Insurance-Grade Auditability for PMFBY

Each 8-day stress and advisory layer is stored with an immutable, timestamped record, creating a continuous satellite-backed evidence trail that can support faster, more objective PMFBY crop-loss claim verification instead of one-off post-damage assessment.

4.8 Zero-License-Cost, Nationally Scalable Stack

Every component — from Sentinel/Landsat/MODIS/Bhuvan data to Google Earth Engine, open-source ML frameworks and free-tier cloud compute — is open or free-tier by design. This removes the licensing-cost barrier that typically blocks pilots from scaling into state- or national-level operational programmes.

5. System Architecture Overview

The pipeline is organized as a sequential, modular flow so that each stage can be independently improved, swapped or scaled without breaking the rest of the system:

Data Ingestion → Pre-processing → Feature Engineering → Crop Classification → Phenology-Aware Stress Detection → Water Deficit & Advisory Engine → Dashboard & Multilingual Delivery

Stage	What Happens	Key Output
Data Ingestion	Pull optical (Sentinel-2/Landsat/MODIS/LISS-III/AWiFS) and SAR (Sentinel-1/EOS-04) tiles for the AOI via GEE / Bhoonidhi / Bhuvan APIs.	Raw multi-temporal stack
Pre-processing	Atmospheric correction, cloud masking, Refined-Lee speckle filtering, 8-day/weekly temporal compositing.	Clean analysis-ready data
Feature Engineering	NDVI, EVI, NDWI, VCI, SAR VV/VH and VH/VV ratio, GLCM texture, phenometrics (SOS, peak, LGP).	Feature cube per pixel
Crop Classification	Foundation-model embeddings + Random Forest / XGBoost on multi-temporal features.	Seasonal crop-type map
Stress Detection	Stage-aware fusion of NDVI anomaly, NDWI, SAR sensitivity into VCI / SMI indices.	Stage-wise stress map
Water Deficit & Advisory	FAO-56 crop-coefficient ET _c estimation vs. rainfall + actual ET; deficit thresholds drive advisory classes.	8-day deficit & advisory map
Dashboard & Delivery	React/Mapbox dashboard, API layer, and multilingual SMS/WhatsApp/IVR push to end users.	Actionable advisory in the field

6. Technology Stack (Latest)

Layer	Technologies
Optical Satellite Data	Sentinel-2 L2A, Landsat-8/9 OLI-2, MODIS MOD13Q1, ISRO LISS-III / AWiFS (via Bhuvan / Bhoonidhi)
Microwave SAR Data	Sentinel-1 GRD/SLC, ISRO EOS-04 SAR, NISAR (planned, modular adapter)
Ancillary / Met Data	IMD gridded rainfall, ERA5 / CHIRPS reference ET, FAO CROPWAT / AquaCrop crop coefficients, canal command-area GIS layers
Geospatial Processing	Google Earth Engine, Python (Rasterio, GDAL, Xarray, Rioxarray), QGIS for QA
Foundation Models	Google AlphaEarth Foundations, NASA-IBM Prithvi-EO-2.0, Clay Foundation Model (embedding-based few-shot classification)
Classical ML	Random Forest, XGBoost, LightGBM — Scikit-learn
Deep Learning	LSTM / Temporal Fusion Transformer (time-series dynamics), U-Net / fine-tuned Segment Anything Model (spatial segmentation) — PyTorch, TensorFlow
Explainability & Uncertainty	SHAP / LIME for feature attribution, Monte Carlo Dropout / quantile regression for confidence scoring
Backend / API	FastAPI (Python), PostgreSQL + PostGIS for spatial storage, Redis for caching
Orchestration	Apache Airflow / Prefect for scheduled 8-day pipeline runs
Frontend / Dashboard	React + Next.js, Mapbox GL JS / deck.gl, Kepler.gl for time-series geospatial views, Tailwind CSS
Advisory Delivery	LLM-based natural-language generation (multilingual), Twilio / Gupshup APIs for SMS, WhatsApp and IVR voice push
DevOps / Cloud	Docker, GitHub Actions CI/CD, Google Cloud Run / AWS Lambda (serverless batch jobs), Google Colab for rapid prototyping
Mobile / Offline Access	Flutter-based PWA with offline tile caching for low-connectivity rural use

7. Datasets & Data Sources

7.1 Optical Satellite Data

- Sentinel-2 L2A and Landsat-8/9 surface reflectance for fine-resolution vegetation indices.
- MODIS MOD13Q1 for long-term, high-frequency vegetation index time series.

- ISRO LISS-III / AWiFS, sourced through Bhuvan or Bhoonidhi, for indigenous moderate-resolution coverage.

7.2 Microwave SAR Data

- Sentinel-1 GRD (VV/VH) for all-weather backscatter and structure information.
- ISRO EOS-04 SAR for indigenous C-band coverage, with a modular adapter ready for NISAR once available.

7.3 Ancillary & Meteorological Data

- IMD gridded rainfall and ERA5 / CHIRPS reference evapotranspiration data.
- Canal command-area boundaries, soil maps, and FAO crop-coefficient (Kc) tables.

7.4 Ground Truth

- Existing crop-type ground survey points from the pilot study area, used for both model training and accuracy validation.

All of the above sources are accessible through open national (Bhuvan, Bhoonidhi) or international (Copernicus Open Access Hub, USGS EarthExplorer, Google Earth Engine catalog) portals, which keeps the workflow viable both for a 30-hour prototype and for later operational scaling without licensing overhead.

8. Detailed Methodology

8.1 Data Pre-processing

- Atmospheric correction of optical imagery and cloud/shadow masking.
- Refined-Lee speckle filtering on SAR imagery to suppress noise while preserving edges.
- Temporal compositing at weekly, fortnightly and 8-day intervals to build analysis-ready stacks.

8.2 Feature Extraction

- Vegetation indices: NDVI, EVI, NDWI, VCI.
- SAR features: VV, VH backscatter and the VH/VV ratio.
- Texture features via Grey-Level Co-occurrence Matrix (GLCM).
- Phenological metrics: start of season (SOS), peak growth date, length of growing period (LGP).

8.3 Crop Type Classification

- Generate foundation-model embeddings for each pixel/parcel time series.
- Train Random Forest / XGBoost classifiers on embeddings plus hand-crafted multi-temporal features.
- Produce a seasonal crop-type map for the pilot command area.

8.4 Phenology-Aware Moisture Stress Detection

- Compute NDVI anomalies relative to the historical stage-specific baseline.

- Combine NDWI, SAR backscatter sensitivity and soil moisture proxies into a Vegetation Condition Index (VCI) and Soil Moisture Index (SMI).
- Classify stress severity separately for each growth stage rather than using one global threshold.

8.5 Water Deficit Estimation & Irrigation Advisory

- Estimate 8-day crop water demand (ET_c) using the FAO-56 crop-coefficient method ($K_c \times \text{reference ET}$).
- Quantify the deficit between ET_c and the sum of effective rainfall and actual evapotranspiration.
- Translate the deficit into a categorical irrigation-status map (no action / monitor / irrigate soon / irrigate now), each carrying a confidence flag.

8.6 AI/ML Model Development

- Combine optical, SAR and meteorological features into a unified feature store.
- Train LSTM / Temporal Fusion Transformer models for temporal stress-progression prediction, and U-Net / fine-tuned SAM for spatial parcel segmentation.
- Apply SHAP-based explainability so every prediction can be traced back to its driving indices.

8.7 Validation

- Compare crop-type output against held-out ground-truth points using Overall Accuracy and Kappa coefficient.
- Cross-check stress and deficit outputs against any available field reports, weather-station records or known stress events for logical consistency.

9. AI/ML Model Details

Task	Model(s)	Why This Choice
Crop type classification	Foundation-model embeddings + Random Forest / XGBoost	Strong accuracy with minimal labelled ground truth
Temporal stress dynamics	LSTM / Temporal Fusion Transformer	Captures stage-to-stage progression, not just snapshots
Spatial parcel segmentation	U-Net / fine-tuned Segment Anything Model (SAM)	Clean field-boundary-aware outputs at scale
Explainability	SHAP / LIME	Builds operator trust by showing which index drove a decision
Uncertainty quantification	Monte Carlo Dropout / quantile regression	Flags low-confidence advisories instead of presenting false certainty

10. Dashboard & Delivery Layer

The output package is designed to be dashboard-ready from day one of the prototype, with a clear path to field-level delivery:

- Interactive map (Mapbox GL / deck.gl) with togglable crop-type, stress and advisory layers.
- Time-series slider to scrub through 8-day composites and watch stress evolve across a season.
- Field/pixel click-through panel showing the SHAP-based explanation behind a given advisory.
- One-click PDF/CSV export for offline sharing with canal-command officials.
- Multilingual SMS / WhatsApp / IVR push so the same advisory reaches a farmer's phone in plain language, not just an analyst's screen.

11. 30-Hour Hackathon Implementation Plan

Hours	Activities
0–2	Team kickoff, repository scaffold, Google Earth Engine / Bhoonidhi authentication, pilot AOI finalization.
2–6	Build the data-ingestion pipeline for optical (Sentinel-2/Landsat) and SAR (Sentinel-1) tiles over the pilot command area.
6–10	Pre-processing (cloud masking, speckle filtering, compositing) and feature extraction (NDVI/EVI/NDWI, VV/VH, GLCM).
10–14	Train baseline Random Forest / XGBoost crop classifier; experiment with foundation-model embeddings for few-shot gains.
14–18	Build phenology-aware moisture stress indices (VCI/SMI) and stage-wise classification logic.
18–22	Implement FAO-56 ET _c water-deficit estimation and the irrigation-advisory rule engine with confidence flags.
22–26	Build the React/Mapbox dashboard; wire up validation against ground-truth crop points; compute accuracy/Kappa.
26–29	Integrate the multilingual advisory layer demo, polish UX, run end-to-end tests, prepare the pitch deck.
29–30	Final rehearsal and hackathon submission.

12. Evaluation Parameters

- Crop classification accuracy target of greater than 85% Overall Accuracy where representative training/validation samples exist, reported alongside the Kappa coefficient.
- Moisture stress classification must be growth-stage aware and show logical correspondence with the underlying satellite indices (not a flat, season-wide threshold).
- Water-deficit (ETc) estimates benchmarked against FAO-56 reference methodology, tracked via RMSE.
- Irrigation advisory layers judged on credibility for command-area planning — consistency with any available rainfall, weather-station or field-report data.

13. Expected Outcomes & Deliverables

- A working pipeline demonstrated on a defined pilot command area, using pre-identified crop labels and cloud-hosted satellite collections within the 30-hour window.
- A methodology and user interface for generating automated, high-resolution crop-type, stage-wise moisture-stress and irrigation-advisory maps.
- A dashboard-ready package: colour-coded stress maps, growth-stage interpretation, field/pixel-level advisory layers and time-series visualization.
- A scalable model design that can extend from a single pilot command area to larger agricultural regions and multiple crop seasons.

14. Future Scope & Scalability

- Pan-India scale-up by replicating the pipeline across additional canal command areas and agro-climatic zones.
- Direct ingestion of NISAR SAR data once available, for finer-resolution all-weather monitoring.
- IoT soil-moisture sensor fusion to ground-calibrate the satellite-derived stress index in near-real time.
- Drone-based spot ground-truthing to continuously refine classifier training data with minimal manual survey effort.
- Deeper integration with PMFBY claim workflows and the Digital Agriculture Mission's farmer registry for direct benefit delivery.

15. Team MeraCodeLikhDo

Name	Suggested Focus Area
Gaurav Tiwari (Leader)	Team leadership, system architecture and end-to-end pipeline integration

Shubham Singh	Satellite data engineering — optical/SAR ingestion, pre-processing and feature extraction
Prajwal Singh	AI/ML engineering — crop classification, stress detection and water-deficit modelling
Krishna Gupta	Full-stack development — dashboard, API layer and multilingual advisory delivery

Roles above are suggested for execution clarity and can be adjusted based on individual team strengths.

16. Conclusion

This project converts a well-known but operationally difficult challenge — turning raw, multi-sensor satellite data into a trustworthy, field-actionable irrigation advisory — into a single, modular AI pipeline. By fusing optical and SAR data at the phenological-stage level, layering in foundation-model embeddings for few-shot crop classification, and closing the loop with a confidence-aware, multilingual advisory delivery layer, the solution is designed to be both hackathon-demoable in 30 hours and credible as a path toward real operational deployment under programmes like PMKSY, the Digital Agriculture Mission and PMFBY.

Team MeraCodeLikhDo — Gaurav Tiwari, Shubham Singh, Prajwal Singh and Krishna Gupta — aims to demonstrate this end-to-end, from satellite pixel to farmer advisory.